

Assignment 07: MAC Layer 2 Report

1. MANUAL DECODING TABLE

Field	F1	F3
Frame Control binary	00001000 00000001	10110100 00000000
Type / Subtype	Data / Data (0000)	Control / RTS (1011)
ToDS, FromDS	1, 0	0, 0
Other flags set	ToDS only	None
Duration (μs)	44	350
Address 1 (role)	A4:83:E7:11:22:33 (RA)	A4:83:E7:11:22:33 (RA)
Address 2 (role)	00:50:56:AA:BB:CC (TA/SA)	00:50:56:AA:BB:CC (TA)
Address 3 (role)	00:1A:2B:3C:4D:5E (DA)	Not present
Sequence no. / fragment no.	467 / 0	Not present

2. QUESTIONS

Q1. F3, F4 and F5 are one exchange. Name the sender and receiver of each, and explain why the Duration drops from F3 to F4 to F5.

Answer:

F3 (RTS): Sender (TA) = 00:50:56:AA:BB:CC, Receiver (RA) = A4:83:E7:11:22:33, Duration = 350 μs.

F4 (CTS): Sender = A4:83:E7:11:22:33, Receiver = 00:50:56:AA:BB:CC, Duration = 324 μs.

F5 (ACK): Sender = A4:83:E7:11:22:33, Receiver = 00:50:56:AA:BB:CC, Duration = 0 μs.

The duration field reserves the medium for the remaining time required to complete the exchange. As time passes and parts of the frame transmission occur, the remaining required time decreases, which causes the Duration value to drop from F3 to F4 to F5.

By the time F4 was transmitted, the F3 has already been transmitted its time is deducted.

At last the when the ACK was sent, now no other frame is going to transmit after it so time is 0.

Q2. A station that hears only F4 sets its NAV. For how many microseconds, and which slide mechanism does this implement?

Answer:

F4 has Duration = 324 μs Therefore, a station that hears F4 sets its NAV for 324 μs. This implements virtual carrier sensing using the Network Allocation Vector (NAV).

The station does not need to physically detect the transmitter's signal to know that the medium is reserved. It uses the Duration information in the frame to defer transmission.

Q3. F2 has the Retry flag set. How does the receiver use the Retry bit together with the sequence number to discard duplicates?

Answer:

F2 has: **Retry = 1 | Sequence Number = 468 | Fragment Number = 0**

The receiver keeps track of previously received frames using their sequence and fragment numbers, If a frame arrives with: Retry = 1 and the same Sequence Number and Fragment Number as a frame that was already successfully received, the receiver recognizes it as a duplicate retransmission.

Therefore, the duplicate frame can be discarded instead of delivering the same data to the upper layer twice. The receiver checks whether sequence 468, fragment 0 has already been received.

Q4. Why does F6 need a fourth address? Sketch the path it takes, using the WDS figure from the slides.

Answer:

F6 has: **ToDS = 1 | FromDS = 1**

When both bits are set, the frame is travelling through a Wireless Distribution System (WDS). The normal three-address format is not sufficient because the frame must preserve both, the final destination, and the original source.

Therefore, a fourth address is required.

For F6:

RA = 02:11:22:33:44:55

TA = 02:66:77:88:99:AA

DA = 00:50:56:AA:BB:CC

SA = A4:83:E7:11:22:33

The general path is:

Original Source

|
| SA
v

Wireless Distribution System

|
| wireless bridge/link
v

Intermediate AP / WDS node

|
| TA / RA information
v

Destination

|
| DA
v

Final Receiver

The fourth address allows the original source address to be retained while the transmitter and receiver addresses describe the current wireless hop.

802.11 Protocol Standards

1. Frame Control Field (2 Bytes)

First Byte:

Bits	Meaning	Values
0–1	Protocol Version	Normally 00
2–3	Type	00 = Management, 01 = Control, 10 = Data
4–7	Subtype	Depends on the frame type

Control Frame Subtypes

Subtype bits	Frame	Address Details
1011	RTS	Contains two addresses
1100	CTS	Contains one address
1101	ACK	Contains one address

Second Byte:

Bit	Flag	Meaning
0	ToDS	Frame is going toward the Distribution System
1	FromDS	Frame is coming from the Distribution System
2	More Frag	More fragments of the frame follow
3	Retry	Frame is a retransmission
4	Power Mgmt	Station's power-management state
5	More Data	More buffered data is available
6	Protected	Frame body is protected/encrypted
7	Order	Strict ordering requested

2. DURATION/ID (2 BYTES)

For the frames in this assignment, it represents the duration in microseconds for which the medium is reserved. The field is stored in little-endian order.

For example: 2C 00 is 0x002C = 44 μ s.

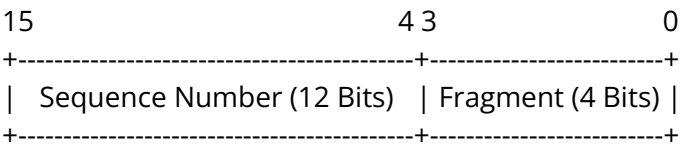
3. ADDRESS FIELDS

802.11 frames can contain different numbers of addresses depending on the frame type and the ToDS/FromDS bits.

The important address roles are:

- RA = Receiver Address
- TA = Transmitter Address
- DA = Destination Address
- SA = Source Address
- BSSID = Basic Service Set Identifier

4. SEQUENCE CONTROL (2 BYTES)



The Sequence Control field is also stored in little-endian order.
Example: F1 contains 30 1D which gives 0x1D30 = 0001 1101 0011 0000
Sequence Number = 0001 1101 0011 = 467 | Fragment Number = 0000 = 0

Frames Summery:

Frame	Type	Subtype	ToDS	FromDS	Other Flags	Duration	Addresses	Seq / Freq	EtherType
F1	Data	Data	1	0		44	3	467 / 0	0x0800
F2	Data	Data	0	1	Retry	44	3	468 / 0	0x0800
F3	Control	RTS	0	0		350	2		
F4	Control	CTS	0	0		324	1		
F5	Control	ACK	0	0		0	1		
F6	Data	Data	1	1	More Frag	44	4	469 / 1	0x0800